

Vibration Analysis Report

Shanghai Metro Line 17 (L17)

Train 1701 / Car 170022 / Position 21

Data Date: 2026-05-28 | 36 Records | 4096 pts/record

Includes: RMS time series, waveform comparison, FFT spectrum analysis

Data Summary

Records: 36 Samples/record: 4096
Time range: 08:12:34 – 13:02:34
Sampling freq: 1293 – 4033 Hz (mean ~2879 Hz)
RMS range: 0.234 – 10.009 g
Peak range: 1.70 – 34.62 g

Key Findings

1. Highest RMS: 10.01 g at 09:29:53
2. Highest peak: 34.62 g at 09:29:58
3. 09:29 cluster: 4 records within 16 s, same section
4. Dominant frequency ~517-534 Hz in harsh events
5. Quiet baseline RMS ~0.23 g (09:05:26)
6. Afternoon (after 12:00) generally lower vibration

Threshold Reference

RMS < 0.5 g : Quiet / stable
RMS 0.5-2 g : Normal operation
RMS >= 2 g : Elevated / event-level

Fig 1. 36 Records RMS Time Series

SH L17 1701-170022 Pos21 - 36 Records RMS Time Series

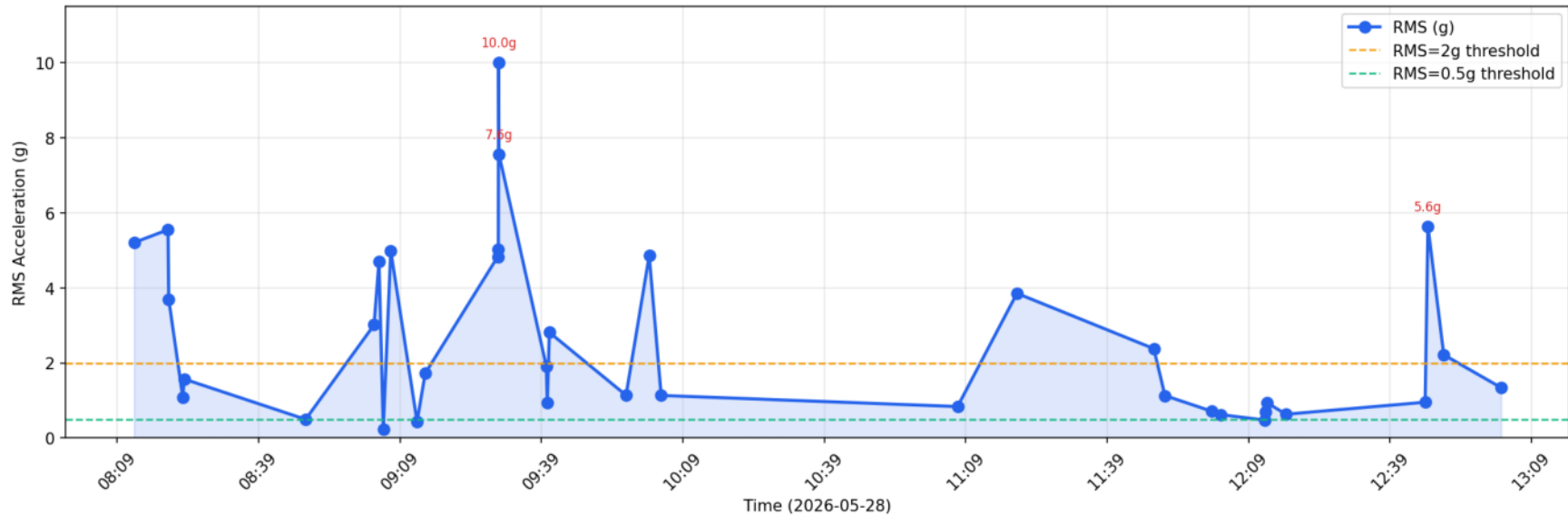


Fig 2. RMS vs Peak Time Series

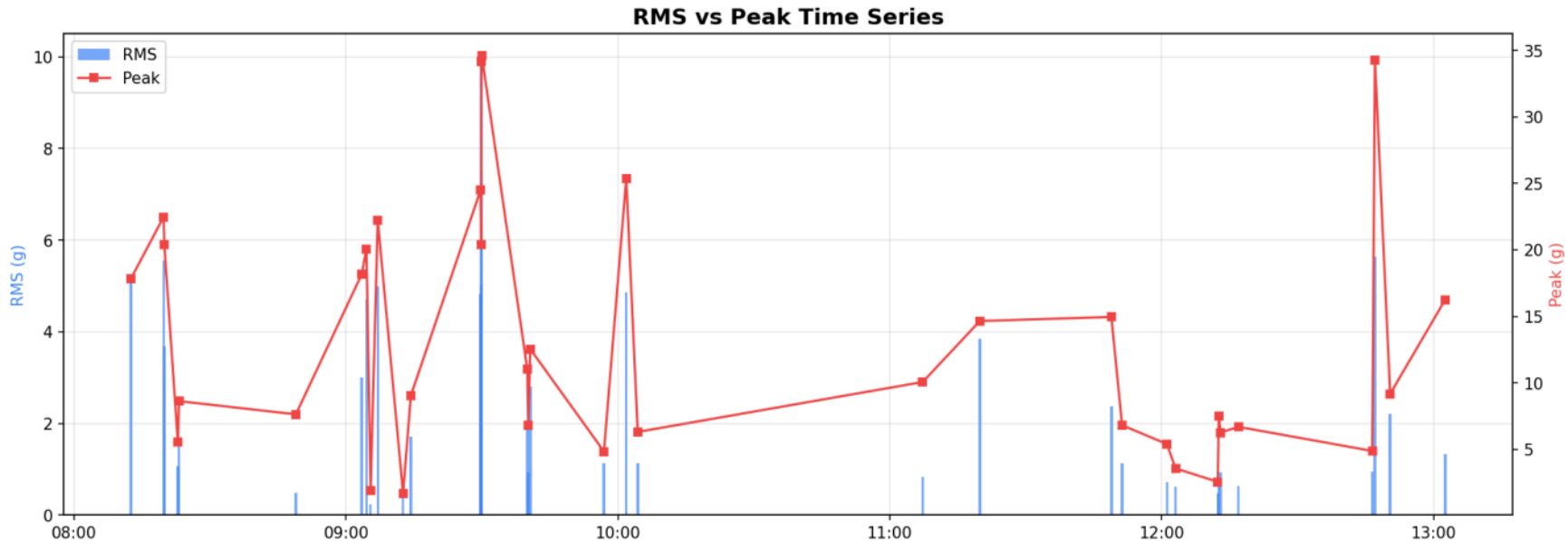


Fig 3. Dominant Frequency vs Time

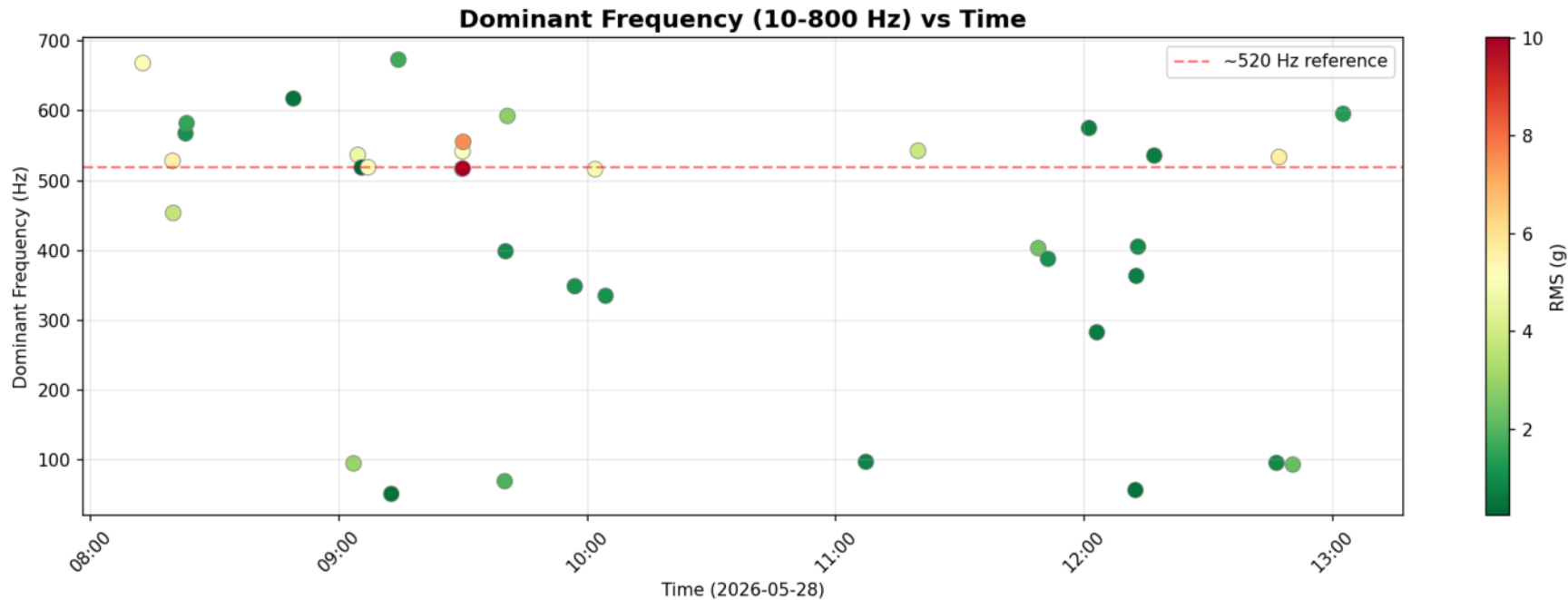


Fig 4. Waveform Comparison (Quiet / Medium / Harsh)

Waveform Comparison by Condition

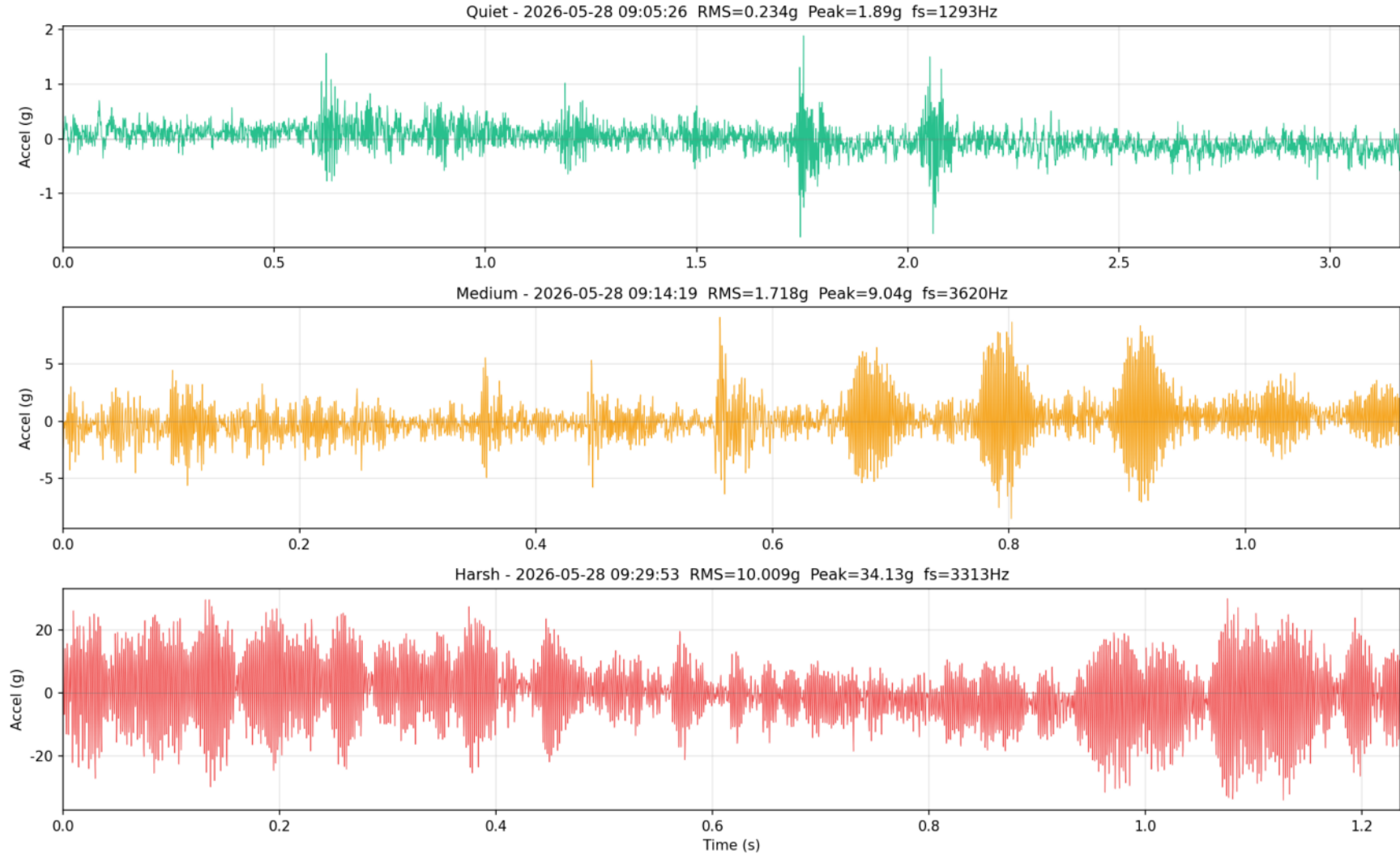


Fig 5. First 0.5s Waveform Overlay

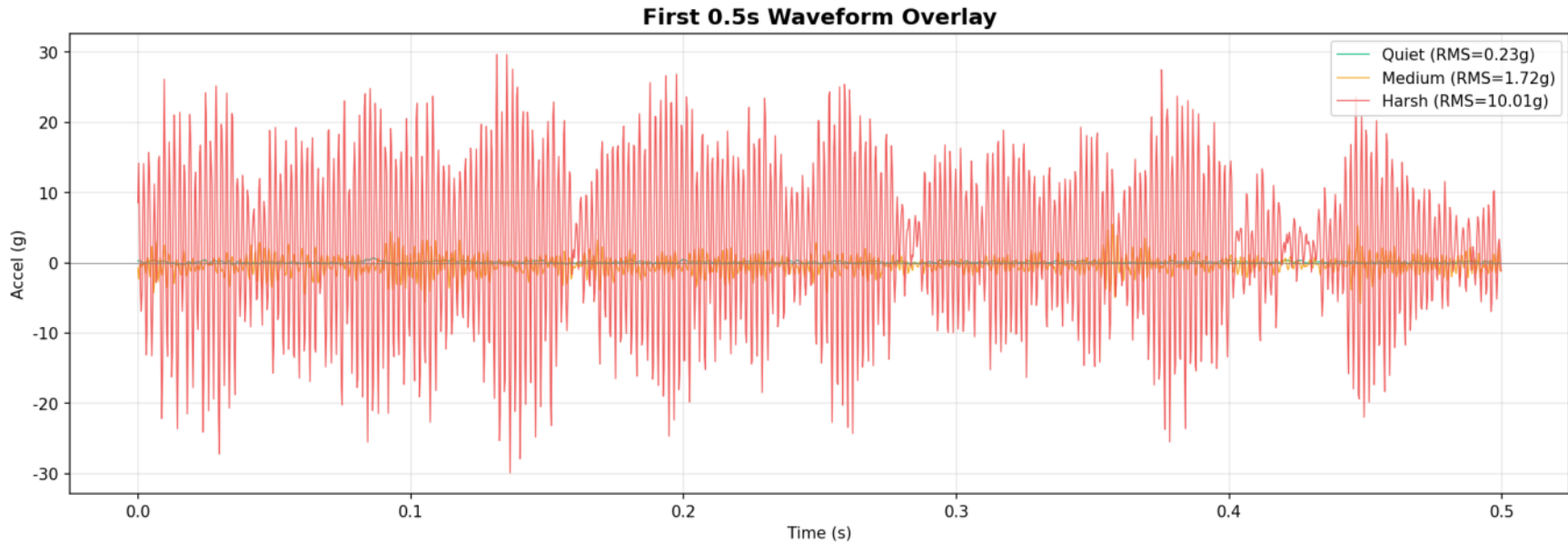


Fig 6. 09:29 High-Vibration Cluster Waveforms

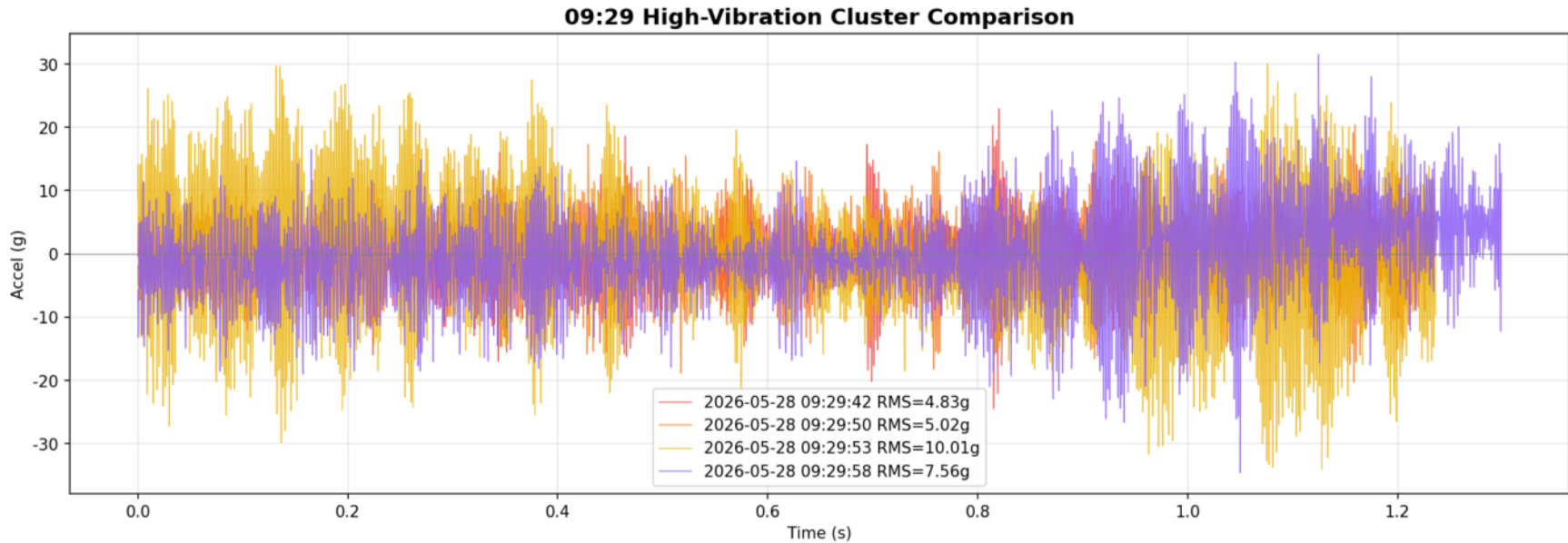


Fig 7. All 36 Waveforms Overview

All 36 Waveforms (color by RMS)

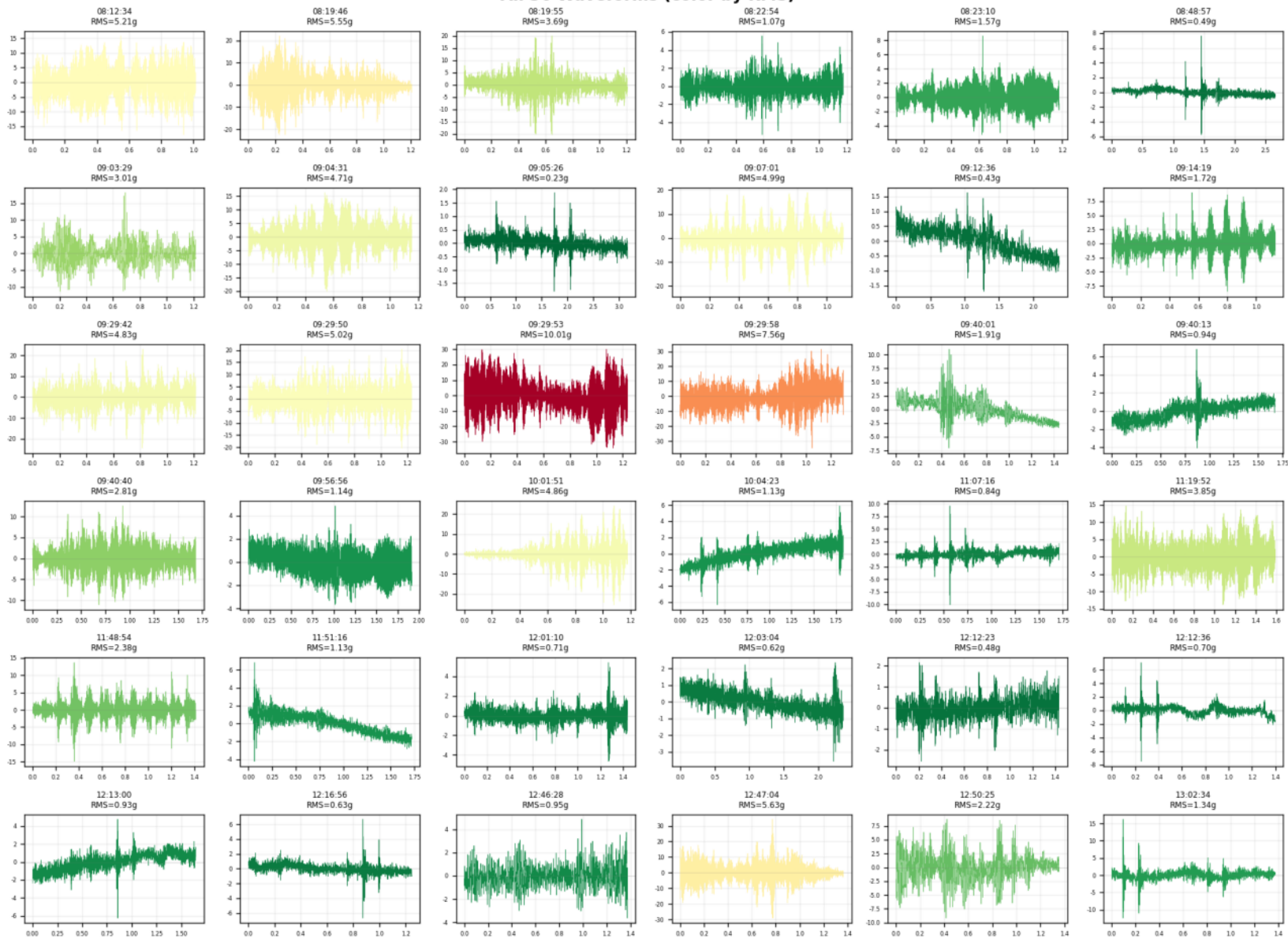


Fig 8. Spectrum Comparison (0-800 Hz)

Spectrum Comparison by Condition (0-800 Hz)

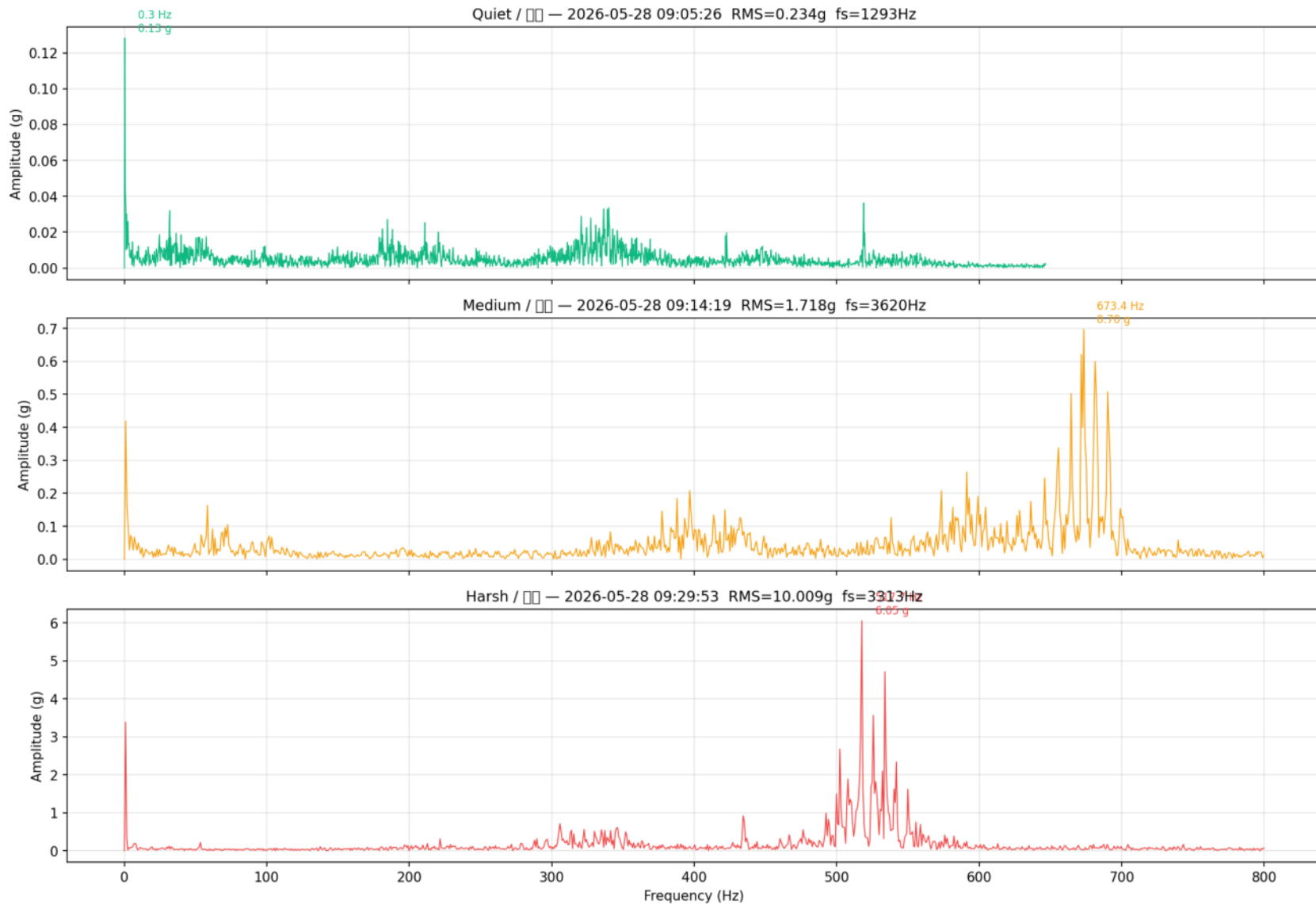


Fig 9. Spectrum Overlay (0-800 Hz)

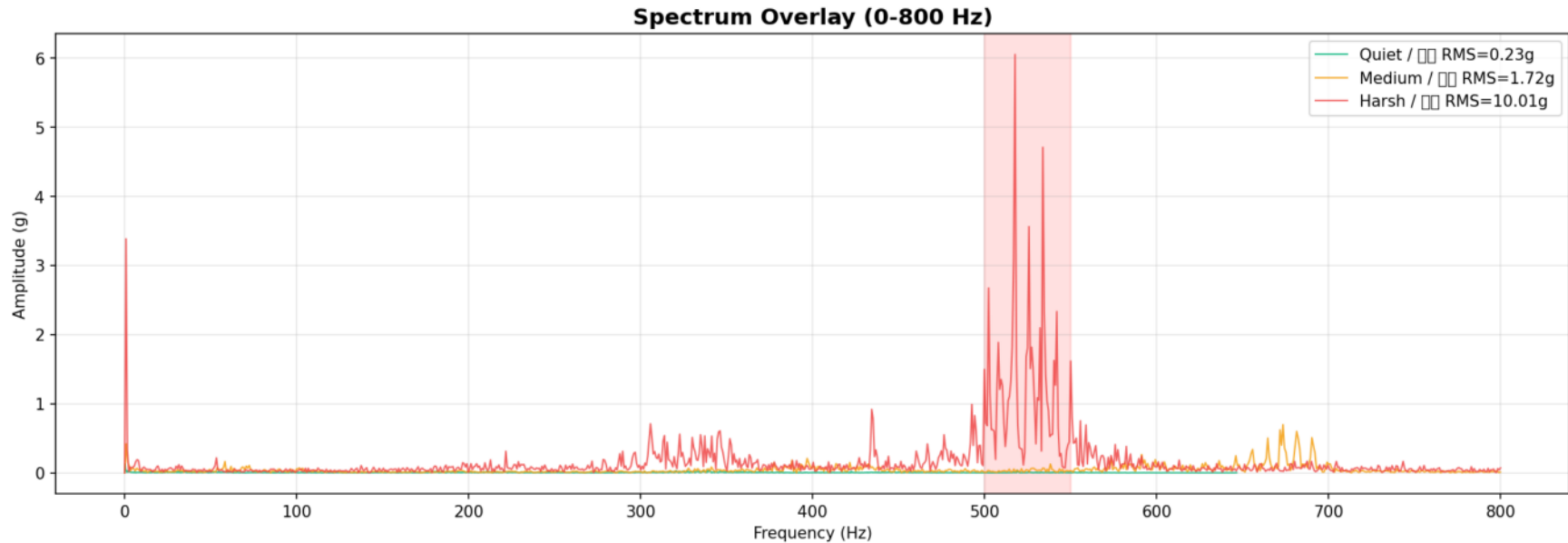


Fig 10. Low-Frequency Spectrum (0-50 Hz)

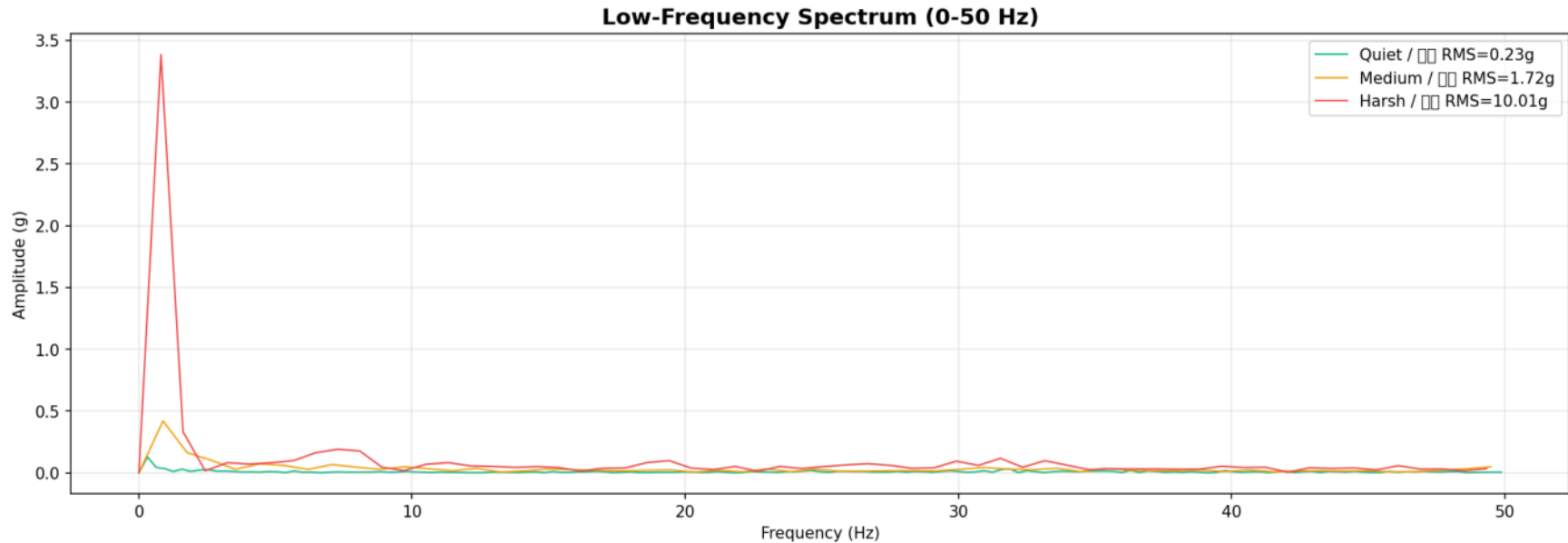


Fig 11. 09:29 Cluster Spectrum Comparison

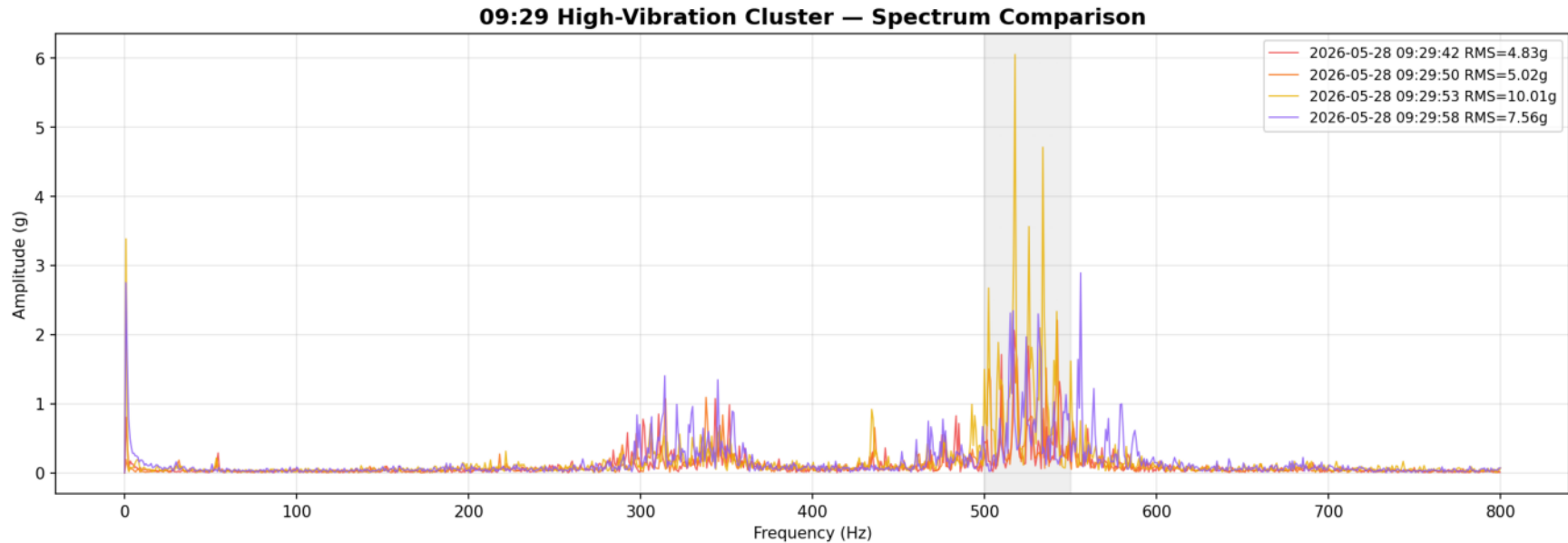


Fig 12. All 36 Spectra Overview

All 36 Spectra (0-800 Hz, color by RMS)



Conclusions & Recommendations

1. Temporal pattern

Vibration is intermittent. ~44% of records have $\text{RMS} \geq 2 \text{ g}$.
The 09:29 time window shows the most severe events.

2. Spectral pattern

Harsh events show strong narrowband energy at ~520 Hz, suggesting wheel-rail coupling or track defect excitation.
Quiet events are dominated by low-frequency ($< 5 \text{ Hz}$) content.

3. Recommended actions

- Cross-reference 09:29 & 12:47 timestamps with train log (speed, location, switch/curve passage)
- Compare ~520 Hz with wheel-pass frequency at that speed
- Use 09:05:26 record as healthy baseline for trending
- Set alert thresholds: $\text{RMS} > 5 \text{ g}$ or $\text{peak} > 25 \text{ g}$ as warning